Institute of Mathematical Sciences
University of Malaya
SIM1001 Basic Mathematics
SIX1013 Fundamentals of Advanced Mathematics

TUTORIAL 3 (Part II)

1. Use elementary row or column operations to find the determinant of the following matrices.

(a) $\begin{bmatrix} 1 & 2 & 8 \\ 6 & 1 & 3 \\ -1 & 5 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 6 & 3 \\ -1 & 5 & 2 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & -1 & 2 \\ 2 & 5 & -3 \\ 4 & 1 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 3 & 4 \\ 5 & 2 & 7 & 8 \\ 1 & 2 & 3 & 0 \\ 0 & 1 & 9 & 4 \end{bmatrix}$.

2. Given that $A = \begin{bmatrix} a & b & c \\ 0 & b & c \\ 0 & 0 & c \end{bmatrix}$.

(a) Determine the values of a, b and c for which A is not invertible.

(b) Determine the values of a, b and c for which A is invertible and find the inverse of A.

3. Let $A \in M_n(\mathbb{R})$ such that $\det A = 9$. Find

(a) $\det(3A)$ (b) $\det(A^{-1})$ (c) $\det(5A^2)$

(d) $\det((5A)^{-1})$ (e) $\det((A^{-1})^T)$

4. Show that

(a) $\det\begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{bmatrix} = (a-b)(b-c)(c-a)$ (b) $\det\begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ b+c & a+c & a+b \end{bmatrix} = 0.$

5. Given that $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 3 \end{bmatrix}$.

(a) Find $\det A$, $\text{adj } A$ and verify $A(\text{adj } A) = (\det A)\, I_3$.

(b) Is A invertible? If so, use $\text{adj } A$ to find the inverse of A.

6. Given the system of linear equations

$$\begin{aligned} x + y + z &= 2, \\ 3x + y &= 1, \\ -x - 2y - 2z &= 3. \end{aligned} \qquad \text{(I)}$$

(a) Compute the determinant of the coefficient matrix of Linear System (I).

(b) Show that the coefficient matrix of Linear System (I) is invertible, and find the inverse.

(c) Solve Linear System (I) by using result (b).

(d) Solve Linear System (I) by using Cramer's rule.

7. Solve the following linear systems using Cramer's rule.

(a) $\begin{aligned} x - y &= 0, \\ y - z &= -1, \\ x + z &= 2. \end{aligned}$ (b) $\begin{aligned} x + y + z &= -1, \\ 2x - y - z &= 4, \\ x + 2y - 3z &= 7. \end{aligned}$

8. For each of the following sets, determine whether it is linearly dependent or linearly independent.

(a) $\{(1, 2, -1), (3, 2, 5)\}$ (b) $\{(1, 3, 3), (0, 1, 4), (5, 6, 3), (7, 2, -1)\}$

9. Let $\boldsymbol{u_1} = (a, 1, -1)$, $\boldsymbol{u_2} = (-1, a, 1)$ and $\boldsymbol{u_3} = (1, -1, a)$ be vectors in $\mathbb{R}^3$. Determine the values of a such that $\boldsymbol{u_1}$, $\boldsymbol{u_2}$ and $\boldsymbol{u_3}$ are linearly independent.

10. Let $\boldsymbol{u_1}$, $\boldsymbol{u_2}$ and $\boldsymbol{u_3}$ be vectors in $\mathbb{R}^n$.

(a) Show that $\boldsymbol{u_1}$, $\boldsymbol{u_2}$ and $\boldsymbol{u_3}$ are linearly independent if and only if $\boldsymbol{u_1}$, $\boldsymbol{u_1}+\boldsymbol{u_2}$ and $\boldsymbol{u_1}+\boldsymbol{u_2}+\boldsymbol{u_3}$ are linearly independent.

(b) Let α, β and γ are nonzero scalars in $\mathbb{R}$. Show that $\boldsymbol{u_1}$, $\boldsymbol{u_2}$ and $\boldsymbol{u_3}$ are linearly independent if and only if $\alpha\boldsymbol{u_1}$, $\beta\boldsymbol{u_2}$ and $\gamma\boldsymbol{u_3}$ are linearly independent.

11. Find the rank of the following matrices:

(a) $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (c) $C = \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix}$

(d) $D = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 0 & -1 & 2 & 3 \end{bmatrix}$ (e) $E = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ (f) $E = \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix}$

12. For each of the following matrices A_i, $i = 1, 2, 3$, find all the values for α such that
(i) $\rho(A_i) = 1$ (ii) $\rho(A_i) = 2$.

(a) $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & \alpha \end{bmatrix}$ (b) $A_2 = \begin{bmatrix} \alpha & 1 \\ 1 & \alpha \end{bmatrix}$ (c) $A_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha^2 \end{bmatrix}$